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Instruction

What the question is

Pumping Lemma Practice: 1.29a&b, 1.30, 1.46a-c, 1.47

1.29 Use the pumping lemma to show that the following languages are not regular.

[A]**a.** $A_1 = \{0^n 1^n 2^n \mid n \geq 0\}$

b. $A_2 = \{www \mid w \in \{a, b\}^*\}$

[redacted]

1.29 a) $A_1 = \{0^n 1^n 2^n \mid n \geq 0\}$ A = reg lang

w/ $p \geq 0$ There # $p \to$ or pumping leng) where $s \to A$ of length at least p s/3 so $s = xyz$

Conditions $i \geq 0$, $xy^iz \in A$ 2. $|xy| \leq p$ 3. $|y| > 0$ p is pumping length so $s = 0^p 1^p 2^p \in A$

$|s| > p$ $|xy| \leq p$, $|y| > 0$. $001122 \to \frac{0}{x} \cdot \frac{0}{y} = \frac{1122}{z}$

Pump mid part

$$\frac{0}{x}\ \frac{0}{y} = \frac{1122}{z} = \frac{0}{x} \cdot \frac{0}{y}\ \frac{1122}{z} * \; i = 2 *$$

$0001122 \neq A_1$ ~~Lang~~ = contradiction

A_1 = Not a regular language.

Double check:

$s = 0^p 1^p 2^p \in A_1$ $|s| > p$ $s = 0^p 1^p 2^p = xyz$ $|xy| \leq p$, $|y| > 0$

001122 A_1; $s = 0^p 1^p 2^p \leq 001122$

$0 = x$ $0 = y$ $1122 = z$

$s = 001122$

$\frac{0}{x}\ \frac{0}{y}\ \frac{1122}{z}$ $xy^iz (i \geq 0)$ $i = 2$ $y \to 00$ thus 0001122

$s = (0)(0)^i(1122)$

$\frac{0}{x}\ \frac{0}{y}\ \frac{1122}{z}$ [when $i = 2$] $0001122 \notin A_1$ $\leftarrow$ Not a reg Lang

1.29 b) $A_2 = \{www \mid w \in \{a, b\}^*\}$ A = reg. Lang.

p = Pumping length by the lemma $s = a^p b a^p b a^p b \in A_2$ $xyz \to |xy| \leq p$

$y > 0$. Thus, $xy^iz \in A_2 \; \forall i \geq 0$

$s = a^p b a^p b = xyz$

make aabaab become string of A_2. Pumping length is 2, $y = a$,

$x = a$ $\to 2x$

s = aabaab $\frac{a}{x}\ \frac{a}{y_2}\ \frac{\text{baabaab}}{z}$ $\leftarrow$ Pump it up $xy^iz (i \geq 0)$ z = baababa baax

$s = (a)(a)(\text{baabaab})$. aaabaabaab $\neq A_2$

A_2 = Not a regular Language.

$s = a^p b a^p b a^p b \in A_2$

xyz

$s = a^p b a^p b a^p b = xyz$ $|xy| \leq p$, $|y| > 0$

s = aabaabaab $xy^iz \in A_2 \; \forall i \geq 0$

$\frac{a}{x}\ \frac{a}{y} = \frac{\text{baabaab}}{z}$ $\frac{a}{x}\ \frac{aa}{y}\ \frac{\text{baabaab}}{z}$ [$i = 2$]

$s = (a)(a)^i(\text{baabaab})$

Not reg lang.

1.30 Describe the error in the following "proof" that 0^*1^* is not a regular language. (An error must exist because 0^*1^* *is* regular.) The proof is by contradiction. Assume that 0^*1^* is regular. Let p be the pumping length for 0^*1^* given by the pumping lemma. Choose s to be the string 0^p1^p. You know that s is a member of 0^*1^*, but Example 1.73 shows that s cannot be pumped. Thus you have a contradiction. So 0^*1^* is not regular.

1.36

P.L = Pumping Lemma, N.R = Not reg

Describe the error that the proof that 0^*1^* is Not Reg Lang.

P.L is utilized as a test usually → to see if the lang is N.R. If P.L shows that meets 1 of the 3 conditions. Also what we talked about in class. then it's N.R. Make $A = lang(0^*1^*)$ & $B = \{0^n1^n | n \geq 0\}$ make/break it into 3 Pieces AKA 0^p1^p $s = xyz$, $i \geq 0$

$0^11^1 = 01$ $0^21^2 = 0011 = 0^31^3 = \underbrace{000}_{x}\underbrace{1}_{y}\underbrace{11}_{z}$ 000111 ~~0001011~~

Y has all 0's than 1 ≠ to lang A. we need to pump it

Z = different Lang. b/c the condit. String Y has all 1's. Pumping String are checked off. The lang. that we talked about in class.

Thus, the proof is that the string can't be a an R by P.L.

Rather, that a Lang that Z can agree w/ the string is N.R.

The above EX: shows the P.L violating 1 condition shows it's N.R.

EX: The P.L is a tool used to prove that a lang is N.R. on the fact if a lang is Reg. Any sufficiently long string in the lang can be "pumped": Have substring repeated to make new strings that also belong to the lang.

2. Proof is incorrect uses a string from a different lang as a counter ex, Specifically, the proof assumes that b/c the string 0^p1^p cannot be pumped in lang $B = \{0^n1^n | n \geq 0\}$, the same conclusion holds for the lang $A = 0^*1^*$.

~~The~~ However, incorrect b/c A & B are diff Lang.

String 0^p1^p is part of A, when you try to apply PL to A w/ this string, the resulting pumped strings ($0^{p+k}1^p$ for some $k > 0$) still belong to A. Thus, A passes the P.L test → cannot use method to show it's N.R

Error is confusing prop of $B = \{0^n1^n\}$ those of $A = 0^*1^*$

Does not show A is non-regular b/c A is indeed reg. The conclusion that 0^*1^* is not R - is incorrect

1.46 Prove that the following languages are not regular. You may use the pumping lemma and the closure of the class of regular languages under union, intersection, and complement.

a. $\{0^n 1^m 0^n \mid m, n \geq 0\}$

A**b.** $\{0^m 1^n \mid m \neq n\}$

c. $\{w \mid w \in \{0,1\}^*$ is not a palindrome$\}$[8]

$\star$ [redacted]

1.46A) $L = \{0^n 1^m 0^n \mid m, n \geq 0\}$

R: L & a string $s = 0^p 1 0^p$ / string into 3 pieces ← Gonna be repeated w/ 2nd & 3rd questions

$s = 0^p 1 0^p = xyz$. p = Pumping length.

$x = 0^{p-k}$, $y = 0^k$ & $z = 10^p$ $(k > 0)$ $xy^0z = 0^{p-k}(0^k)^0 10^p = 0^{p-k}10^p \notin L$

xy^0z Does not belong to L b/c $p-k<p$. $y^0 = \epsilon$

L is regular is a contradiction. Thus, P.L it's proved that L is not reg.

1.46B) $L = \{0^m 1^n \mid m \neq n\}$

L is reg Lang & string $s = 0^p 1^{p+p!}$ / string

$s = 0^p 1^{p+p!} = xyz \in L$ $|s| \geq p$, where $p = P.L$

$p!$ is ÷ by all ints from $1 \to p$, $p! = p \times (p-1) \times (p-2) \times \ldots \times 1$.

$x = 0^a$, $y = 0^b$ & $z = 0^c 1^{p+p!}$, $b \geq 1$ & $a+b+c = p$

$s' = xy^{i+1}z$, where $i = \frac{p!}{b}$ $y^i = 0^{p!}$ $y^{i+1} = 0^{b+p!}$ $xyz = 0^{a+b+c+p!} 1^{p+p!}$ $\notin L$ a

$m = p+p!$, $n = p+p!$ & $m = n$ ← contradiction b/c contradiction.

$m = n$, Thus P.L is proved L is N.R

1.46C $L = \{w \mid w \in \{0,1\}^*$ is not a palidrome$\}$

$\overline{L} = \{w \mid w \in \{0,1\}^*$ is palidrome$\}$ ← complement.

$s = 0^p 1 0^p$. * $s = 0^p 1 0^p = xyz \in L$ $p = P.L$.

$x = 0^{p-k}$, $y = 0^k$ & $z = 10^p$ $k > 0$ $xy^0z = 0^{p-k}(0^k)^0 10^p$

$= 0^{p-k}10^p \notin L$ $(y^0 = \epsilon)$

String xy^0z not same from forward & backward direction b/c $p-k<p$. The string xy^0z D.N. belong to $\overline{L}$. Assumption is a contradiction, P.L is proved = Not Reg.

1.47 Let $\Sigma = \{\mathtt{1}, \mathtt{\#}\}$ and let

$$Y = \{w \mid w = x_1\mathtt{\#}x_2\mathtt{\#}\cdots\mathtt{\#}x_k \text{ for } k \geq 0, \text{ each } x_i \in \mathtt{1}^*, \text{ and } x_i \neq x_j \text{ for } i \neq j\}.$$

Prove that Y is not regular.

1.47 $\Sigma = \{1, \#\}$

The Problem

$x_1 \# x_2 \# ... \# x_k$ where $x_1, x_2, ..., x_k$ x_i = can be # of 1s

Y contains only 1s & #'s b/c {1,#} + input alphabet. ∃ substring ≠ $x_i \neq x_j$.

2 S.S are separated the input alphabet #

A is = R.L. there is a # P (Pumping length) where S is any string that belongs to A of length at least P then P, then S may be divided into 3 pieces S = uvw. 1. For each $i \geq 0$, $uv^iw \in A$ 2. $|v| > 0$ & 3. $|uv| \leq p$

Y = R.L let p = Pump Length for Y. Y = strings of Lang are of the form W = $x_1 \# x_2 \# ... \# x_k$, $S = x_1 \# x_2$. Consider a string $S = x_1 \# x_2$ for k = 2 x_1 ∉ x_2

x_1 & x_2 can be formed w/ only 1s but both ∉. Any 2 differ str can be taken for x_1 & x_2. $x_1 = 1^p$ & $x_2 = 111^p$. Then the str. $S = 1^p \# 11^p$

x_1 & x_2 are different substrings & the val x_1 & x_2 depends on the p val. For EX p = 2 then val x_i & x_2 are 111 & 1111. length of S < than p & S ∈ Y. Allow 111#1111 be the str that belongs to Y. The P. length of the str. = 2, / the str 111#1111 into 3 parts = x, y, z

x = 1 y = 1 w = 1#1111 (the last part of str).

$$S = 111\#1111 = \underset{x}{\underline{1}} \; \underset{y}{\underline{1}} \; \underset{w}{\underline{1\#1111}}$$

& pump middle part xy^iz ($i \geq 0$. For i = 2, v = 11.

$$\underset{u}{\underline{1}} \; \underset{v}{\underline{11}} \; \underset{w}{\underline{1\#1111}} \; (i = 2)$$

1111#1111 ∉ Y b/c the S.S $x_1 = x_2$ which violates condition of the Lang Y. It's contradicting. PL = violated

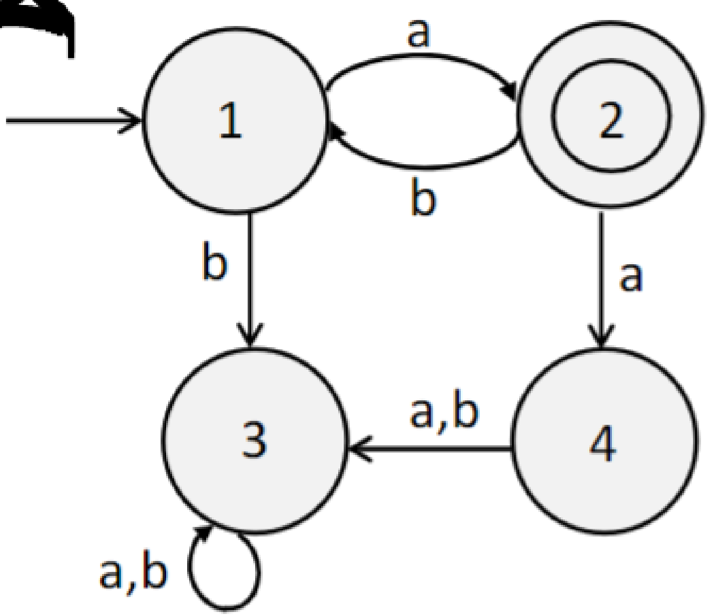
Thus Y is NOT Regular Language

State Minimization Exercises.pdf

State Minimization Exercises

Find the minimum state DFA for each of these machines, and write a simple regular expression for each.

A



Graph A

So we need to use state equiv

0 {1, 4, 3}, {2}

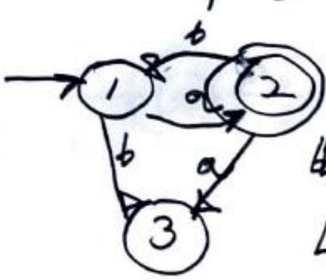
non - final

1 {1}, {3, 4} {2}

{1}, {3, 4}, {2}

They are the same

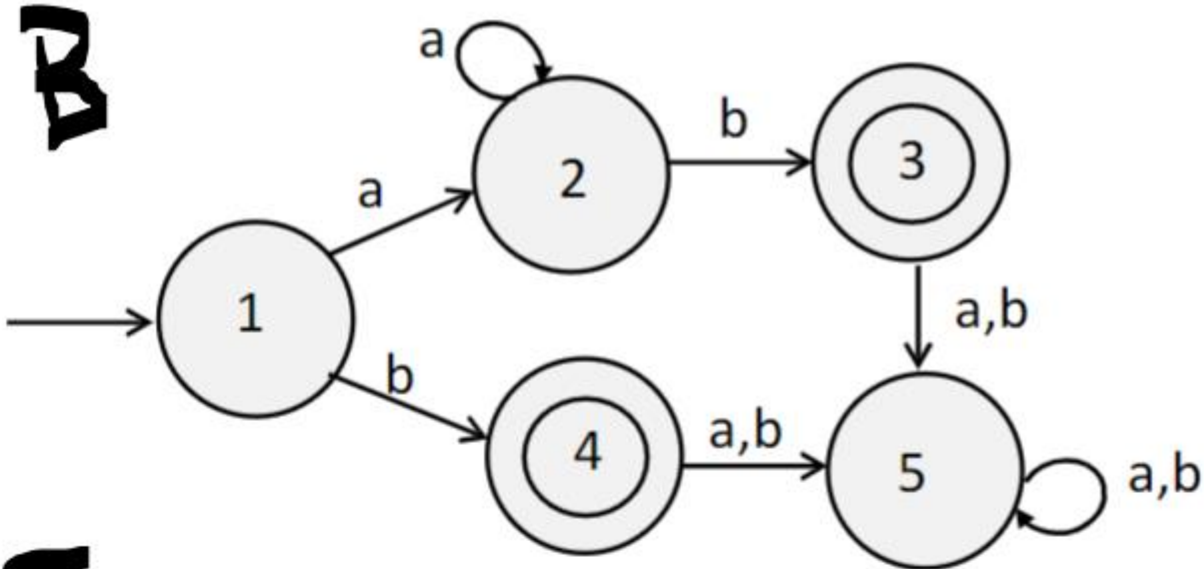
Now lets stop & 3, 4 states so it merges or you merge together



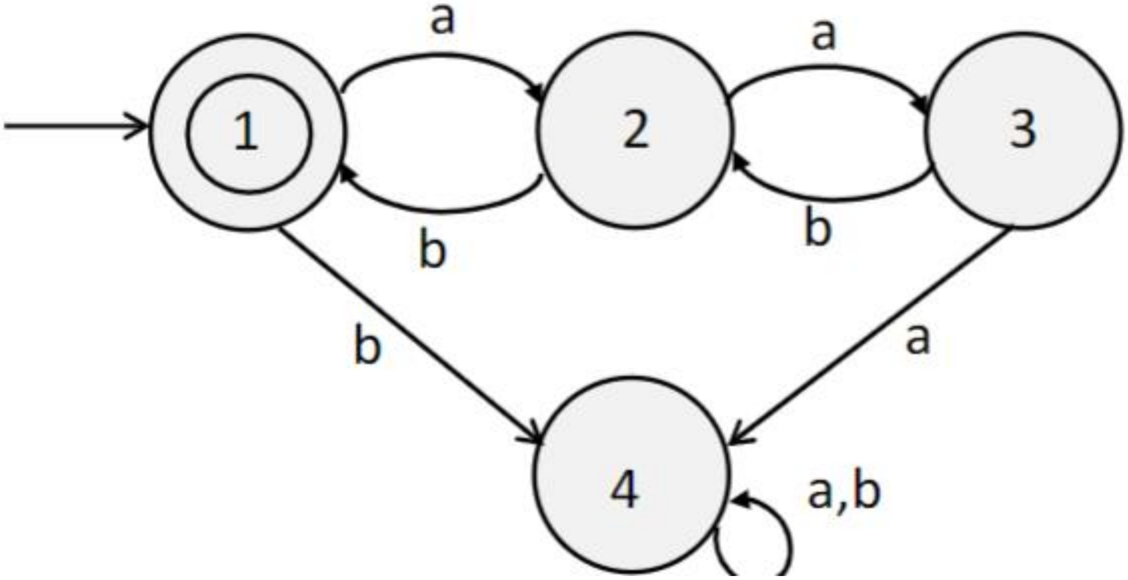
Lemma

$L = a(ba)^*$

B



C

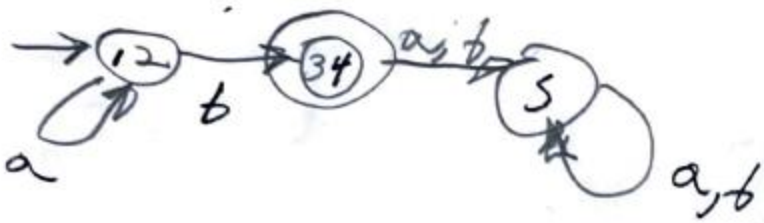


Graph B:

equivalence	nonL :	Final
0	{1,2,5}	, {3,4}
1	{1,2},{5}	, {3,4}
2	{1,2}, {5}	, {3,4}

They are = . So we stop

1,2 & 3 4 are equivalent state



$\boxed{L = a^*b}$

Graph C:

Equivalence

0	{1} {2,3,4}
1	{2},{3 4}
2	{2},{3},{4}

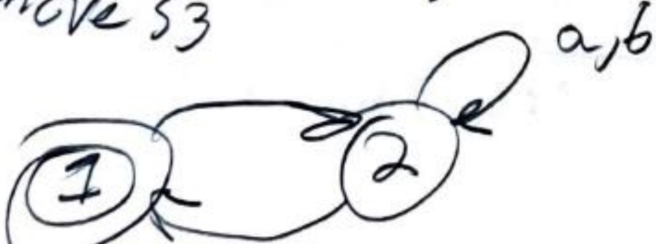
No redundent State. Given DFA is minimal $L=[a(ab)^*b]+\epsilon$

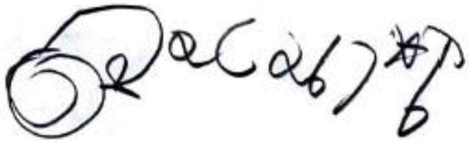
D.C:

D.C can remove 4

S2

remove S3





$(a(ab)^*b)^*$

$2=(\epsilon+ab)a+(2a)b \rightarrow a+2ba+2ab$

$2=a+2(ba+ab)$